



UPY 
**UNIVERSIDAD
POLITÉCNICA
DE YUCATÁN**

AI-Powered study game

PROGRAMMING

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Data and AI Engineering

2nd Cuadrimestre

May-August
2026

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INTRODUCTION

- The Hook: We consume massive amounts of data daily, yet when it comes to studying, we still rely on static, repetitive reading.
- The Context: Rote memorization often leads to burnout and low retention rates among university students.
- The Approach: Gamification has been proven to increase engagement. By leveraging Artificial Intelligence, we can transform any static document into a dynamic, personalized challenge.

OBJECTIVE

To design an AI-powered educational system that processes user-provided study materials and target topics to automatically generate an interactive, game-based learning experience for exam preparation.

PROBLEM STATEMENT

- Students struggle to extract the most important concepts from dense, unstructured academic texts and study materials.
- Creating effective, unbiased self-evaluation tools (like quizzes or flashcards) manually is highly time-consuming.
- There is a lack of accessible tools that instantly convert specific, user-provided topics into interactive learning sessions to test real comprehension.

IPO SPECIFICATION: INPUT

Defining the variables and data types required for system initialization.

Variable Name	Data Type	Example Value
studyTopic	String	"Cell Biology"
sourceMaterialText	String	"Mitochondria are membrane-bound cell organelles..."
gameDifficultyLevel	Integer	3 (representing "Hard")
numberOfQuestions	Integer	10

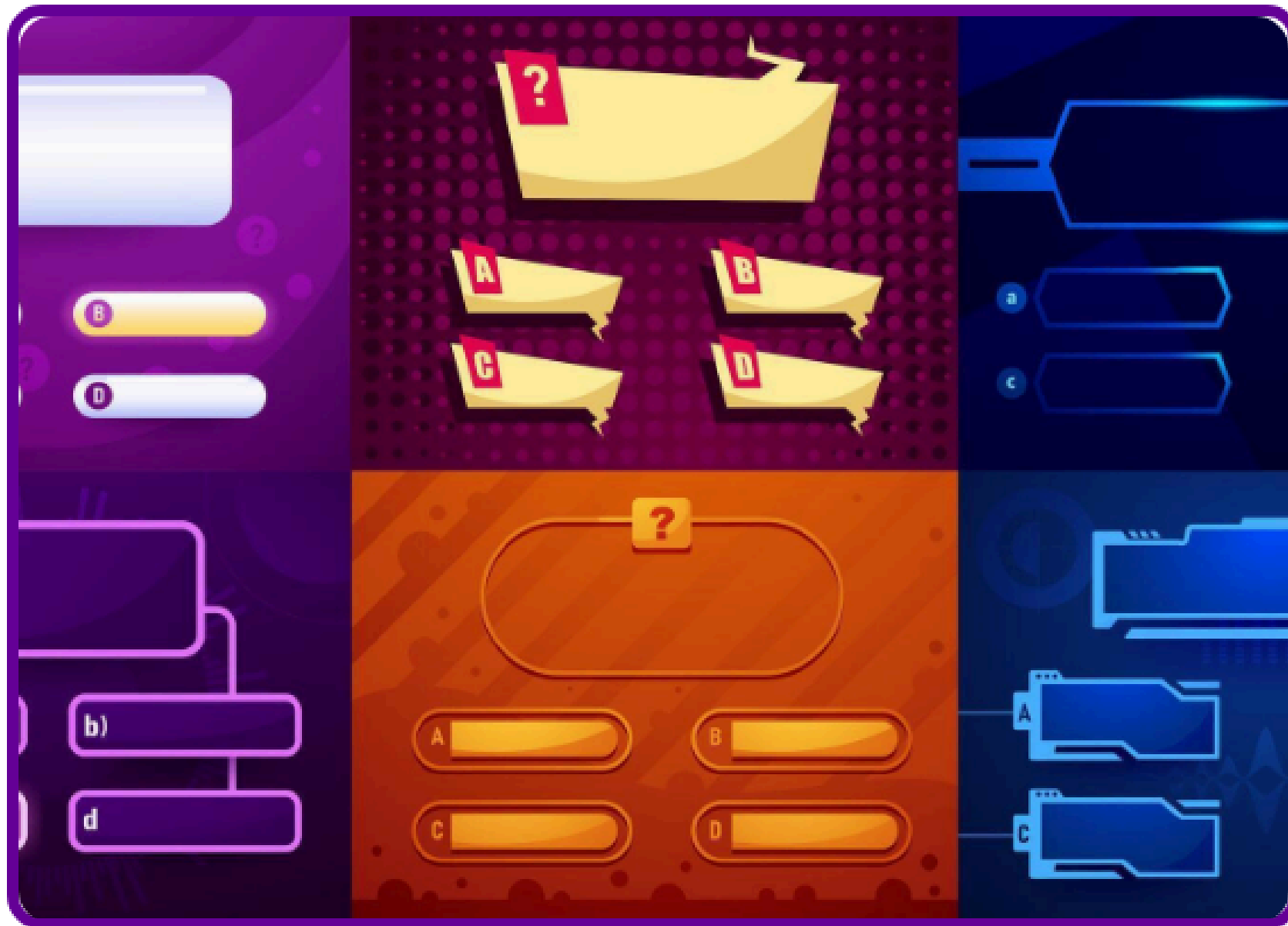
IPO SPECIFICATION: PROCESS

The system logic follows a sequential data pipeline:

1. **Data Retrieval:** Capture text and parameters.
2. **NLP Extraction:** AI identifies core entities.
3. **Heuristic Generation:** Creation of questions and distractors.
4. **Interface Loop:** Interactive Q&A execution.

```
# Algorithmic Logic
READ studyTopic, sourceMaterialText
STORE inputs in session_memory
EXTRACT key_concepts FROM text
GENERATE questions BASED ON difficulty
DISPLAY game_interface
CALCULATE score IF answer IS correct
```

IPO SPECIFICATION: OUTPUT



Concrete System Results:

- **Interactive UI:** Multiple-choice game screen with randomized options.
- **Live Score Tracker:** Real-time numerical feedback after every input.
- **Session Dashboard:** Summary showing accuracy %, study time, and concept mastery.



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THANK YOU!

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APPENDIX

Variable Dictionary

Comprehensive list of secondary variables, memory address markers, and data structures for AI context management.

Flow Diagrams

Visual logic flows for API calls.

Scalability

Future multi-player support.

Security

Data privacy for user notes.

TECHNICAL LOGIC

```
# System Core Loop
WHILE game_active IS TRUE
  DISPLAY question_array[index]
  READ user_input <- from GUI_BUTTON
  IF user_input == correct_answer THEN
    session_points <- session_points + difficulty_multiplier
    DISPLAY "Correct!" with audio_feedback
  ELSE
    DISPLAY "Try again!"
  ENDIF
  index <- index + 1
ENDWHILE

DISPLAY "Study Complete!"
GENERATE report_PDF(session_points, studyTopic)
```